Department of Mathematics and Applied Mathematics

Module 3MS: Metric Spaces

Class Test Two SOLUTIONS **10 May 2018**

Time : 18h00 – 19h30 **Total marks: 40**

1. (a) i. True
   ii. False
   iii. False
   iv. True

   (b) $[0,1] \cup [2,\infty)$ with the metric d_E of course. [5]

2. Let d_s be the metric on $\mathbb{R}^2$ given by $d_s((a,b),(c,d)) = |a-c| + |b-d|$.

   (a) Show that, for all (a,b), $(c,d) \in \mathbb{R}^2$,

$$\frac{1}{2}d_s((a,b),(c,d)) \leq d_E((a,b),(c,d)) \leq d_s((a,b),(c,d))$$

For the first inequality: $\frac{1}{2}d_s((a,b),(c,d)) \leq d_E((a,b),(c,d)) \iff$

$d_s((a,b),(c,d)) \leq 2d_E((a,b),(c,d)) \iff$

$|a-c| + |b-d| \leq 2\sqrt{(a-c)^2 + (b-d)^2}$.

Since $|a-c| \leq \sqrt{(a-c)^2 + (b-d)^2}$ and $|b-d| \leq \sqrt{(a-c)^2 + (b-d)^2}$, we have $|a-c| + |b-d| \leq 2\sqrt{(a-c)^2 + (b-d)^2}$.

For the second inequality: $d_E((a,b),(c,d)) \leq d_s((a,b),(c,d)) \iff$

$\sqrt{(a-c)^2 + (b-d)^2} \leq |a-c| + |b-d| \iff$

$(a-c)^2 + (b-d)^2 \leq (|a-c| + |b-d|)^2 = (a-c)^2 + (b-d)^2 + 2\,|a-c|\,|b-d|$, which holds.

(b) Show that, for any sequence $((a_n, b_n))$ in $\mathbb{R}^2$, $(a_n, b_n) \to (a, b)$ with respect to d_E if and only if $(a_n, b_n) \to (a, b)$ with respect to d_s.

Suppose $(a_n, b_n) \to (a, b)$ with respect to d_E. Then $d_E((a_n, b_n), (a, b)) \to 0$.
Now $0 \leq d_s((a_n, b_n), (a, b)) \leq 2d_E((a_n, b_n), (a, b))$ for all n.
Taking limits as $n \to \infty$ gives $d_s((a_n, b_n), (a, b)) \to 0$, and so $(a_n, b_n) \to (a, b)$ with respect to d_s.

Suppose $(a_n, b_n) \to (a, b)$ with respect to d_s. Then $d_s((a_n, b_n), (a, b)) \to 0$.
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Taking limits as $n \to \infty$ gives $d_E((a_n, b_n), (a, b)) \to 0$, and so $(a_n, b_n) \to (a, b)$ with respect to d_E.

[5]

3. (a) Prove or disprove: Every isometry is one-one.

True.
Suppose that $f : (X, d) \to (Y, \rho)$ is an isometry, so that $\rho(f(a), f(b)) = d(a, b)$ for all $a, b \in X$.
Suppose $f(a) = f(b)$.
Then $\rho(f(a), f(b)) = 0$; so $d(a, b) = 0$.
So $a = b$.

(b) Suppose that $f : X \to Y$ is a continuous function between metric spaces. Prove that, for $A \subseteq X$, $f(\overline{A}) \subseteq \overline{f(A)}$.

Suppose $A \subseteq X$; take $y \in f(\overline{A})$. Then $y = f(x)$ for some $x \in \overline{A}$. Then there exists a sequence (a_n) in A such that $a_n \to x$. Since f is continuous, $f(a_n) \to f(x)$. Since the sequence $(f(a_n))$ is in $f(A)$, this shows that $f(x) \in \overline{f(A)}$, as required.

(c) Let X and Y be sets and $f : X \to Y$ a function. Suppose that $\mathcal{T}$ is a topology on Y. Prove that $\mathcal{U} = \{f^{-1}(G) : G \in \mathcal{T}\}$ is a topology on X.

$\emptyset \in \mathcal{T}$, so $f^{-1}(\emptyset) = \emptyset \in \mathcal{U}$
$Y \in \mathcal{T}$, so $f^{-1}(Y) = X \in \mathcal{U}$
If $G_1, G_2 \in \mathcal{T}$, then $G_1 \cap G_2 \in \mathcal{T}$. But $f^{-1}(G_1) \cap f^{-1}(G_2) = f^{-1}(G_1 \cap G_2)$, so $\mathcal{U}$ is closed under binary intersections.
If $G_i \in \mathcal{T}$ for all $i \in I$, then $\bigcup_{i \in I} G_i \in \mathcal{T}$. But $\bigcup_{i \in I} f^{-1}(G_i) = f^{-1}(\bigcup_{i \in I} G_i)$, so $\mathcal{U}$ is closed under arbitrary unions.

Please note:
(1) It's fine to check closure under binary intersections, instead of closure under finitely many. However, it's not fine to check closure under binary intersections, instead of closure under arbitrarily many.
(2) For an arbitrary index set, please use some letter say I and index over $i \in I$. Do not use natural numbers. (Reason: there may be cases where the indexing set needs to be uncountable.)

[4+4+4=12]

4. **Lemma** Let (x_n) be a Cauchy sequence in a metric space (X, d). Then the diameter of the set $\{x_n : n \in \mathbb{N}\}$ is finite.

Proof Choose $N \in \mathbb{N}$ such that $m \geq n \geq N$ implies that $d(x_n, x_m) < 1$. (1)
Let $M = \max\{d(x_r, x_s) : r = 1, \ldots, N \text{ and } s = 1, \ldots, N\}$. (2)
Then $d(x_p, x_q) \leq M + 1$ for all $p, q \in \mathbb{N}$. (3)
So diam $\{x_n : n \in \mathbb{N}\} \leq M + 1$. (4)

(a) Write down the definition of the diameter of a subset A of a metric space (X, d).

If $A = \emptyset$, diam $A = 0$.
If $A \neq \emptyset$, diam $A = \sup\{d(a, b) : a, b \in A\}$, if this supremum exists; otherwise the diameter of A is infinite, or undefined.

(b) Why do we know that an N as mentioned in line (1) exists?

Because (x_n) is a Cauchy sequence.

(c) In line (2), we take the maximum of a set. How do we know that maximum exists?

The set only has finitely many real numbers in it, so its maximum exists.

(d) Prove the claim in line (3).

For $p, q \leq N$, $d(x_p, x_q) \leq M \leq M + 1$.
For $p, q \geq N$, $d(x_p, x_q) < 1 \leq M + 1$.
For $p \leq N$ and $q \geq N$, $d(x_p, x_q) \leq d(x_p, x_N) + d(x_N, x_q) \leq M + 1$.

(e) For the sequence $\left(\left(\frac{1}{n}, \frac{2}{n}\right)\right)$ in $(\mathbb{R}^2, d_s)$ find the diameter of $\left\{\left(\frac{1}{n}, \frac{2}{n}\right) : n \in \mathbb{N}\right\}$. Give a brief calculation to justify your answer. (See Question 2 above for the definition of d_s.)

$$\text{diam}\left\{\left(\frac{1}{n}, \frac{2}{n}\right) : n \in \mathbb{N}\right\} = \sup\left\{d_s\left((\tfrac{1}{p}, \tfrac{2}{p}), (\tfrac{1}{q}, \tfrac{2}{q})\right) : p, q \in \mathbb{N}\right\}$$

$$= \sup\left\{\left|\tfrac{1}{p} - \tfrac{1}{q}\right| + \left|\tfrac{2}{p} - \tfrac{2}{q}\right| : p, q \in \mathbb{N}\right\} = 3$$

[3+3+2+2+2=12]

5. (a) Prove or disprove: If A is a closed subset of a metric space X, and the boundary of A is a non-empty, complete subset of X, then A is a complete subset of X.

False.
Consider the metric space $(X, d) = ((0, 2), d_E)$ and $A = (0, 1]$.
Here A is a closed subset of X.
The boundary of A is $\{1\}$, which is a non-empty, complete subset of X.
However, A is not complete, because $(\frac{1}{n})$ is a Cauchy sequence in A that does not converge to a point of A.

(b) Prove or disprove: If X and Y are uniformly isomorphic metric spaces, and the diameter of X is finite, then the diameter of Y is finite.

False.
If d is a metric on X, define another metric ρ on X by

$$\rho(x, y) = \frac{d(x, y)}{1 + d(x, y)}$$

Now the identity function id: $(X, d) \to (X, \rho)$ is a uniform isomorphism, because $d(x_n, y_n) \to 0$ iff $\rho(x_n, y_n) \to 0$.
However, ρ is bounded by 1, while d need not be (take d_E on $\mathbb{R}$ for instance).

[2+2=4]

6. Take $x \in Z$. Since $x \in \overline{A}$, there exists (a_n) in A such that $a_n \to x$.

Claim: $(f(a_n))$ is a Cauchy sequence in Y.
Proof: Let $\epsilon > 0$ be given. Choose $r > 0$ such that diam $f[A \cap B(x; r)] < \epsilon$.
Since $a_n \to x$, there exists N such that $n \geq N \implies a_n \in B(x; r)$.
So, for $m, n \geq N$ we have $d(f(a_n), f(a_m)) \leq$ diam $f[A \cap B(x; r)] < \epsilon$.

Define $F(x) = \lim f(a_n)$. This uses completeness of Y.

Claim: F is well-defined.
Proof: Suppose (a_n) and (c_n) are sequences in A with $a_n \to x$ and $c_n \to x$.
Let $y = \lim f(a_n)$ and $y' = \lim f(c_n)$.
Then $d_Y(y, y') = \lim d_Y(f(a_n), f(c_n))$.
Let $\epsilon > 0$ be given. Choose $r > 0$ such that diam $f[A \cap B(x; r)] < \epsilon$.
Choose N such that $n \geq N \implies a_n \in B(x; r)$ and $c_n \in B(x; r)$.
Then $n \geq N \implies d_Y(f(a_n), f(c_n)) < \epsilon$.
So $\lim d_Y(f(a_n), f(c_n)) = 0$ and so $y = y'$.

Claim: F is continuous. Proof: Let $x \in Z$ and $\epsilon > 0$ be given.

Choose $r > 0$ such that diam $f[A \cap B(x;r)] < \frac{\epsilon}{2}$.
Let $\delta = \frac{r}{2}$.
Let $y \in Z$ with $d(x,y) < \frac{r}{2}$. Write $x = \lim a_n$ and $y = \lim b_n$ as before.
Choose N such that $n \geq N \Longrightarrow d(a_n, x) < r$ and $d(b_n, y) < \frac{r}{2}$.
Then, for $n \geq N$, $a_n, b_n \in A \cap B(x;r)$ (where the latter uses $d(b_n, x) \leq d(b_n, y) + d(y, x) < \frac{r}{2} + \frac{r}{2} = r$.)
So, for $n \geq N$, $d_Y(f(a_n), f(b_n)) < \frac{\epsilon}{2}$, giving $d_Y(F(x), F(y)) = \lim d_Y(f(a_n), f(b_n)) < \epsilon$.

[2]